HW-1

1. 23 people, 365 possibilities, equal probability

(a) P(all 23 people have distinct bday)

$P(1^{st}) = 1 = \frac{365}{365}$, $P(2^{nd}) = \frac{364}{365}$

$P(23^{rd}) = \frac{365-22}{364} = \frac{343}{365}$

$P(\text{all distinct}) = 1 \times \frac{364}{365} \times \dots \times \frac{343}{365}$

$= \frac{364 \times \dots \times 343}{365^{22}} \approx 0.4927$

(b) $P(\overbrace{\text{at least 2 people share the bday}}^{\text{event A}})$

$P(A) = 1 - P(\text{all distinct})$

$= 1 - 0.4927 = 0.5073$

(c) This means that there is a 50% chance that among 23 people, at least 2 of them share a bday. We do not compare 1 against 22. We compare every possible pair $({}^{23}C_2) \approx 253$ pairs.

2. $P(D) = 0.01$, $P(D^c) = 0.99$

$P(+|D) = 0.95$

$P(-|D^c) = 0.98 \Rightarrow \underbrace{P(+|D^c)}_{\text{FPR}} = 0.02$

We know test is +ve $\Rightarrow (+)$

(a) $P(D|+) = \frac{P(+|D)\,P(D)}{P(+)}$

$$P(+) = P(+|D)\,P(D) + P(+|D^c)\,P(D^c)$$
$$= (0.95 \times 0.01) + (0.02 \times 0.99)$$
$$= 0.0293$$

$$P(D/+) = \frac{0.0095}{0.0293} = 0.3242$$

(b) $P(D^c|+) = \frac{P(+|D^c)\,P(D^c)}{P(+)}$

the result is false +ve

$$= \frac{0.02 \times 0.99}{0.0293} = 0.6758$$

(c) Only 1% have popula^n has disease. That means there are lot more healthy people than diseased ∴, even a small 2% FPR on 99% healthy popula^n becomes a large no. of false positives

3. 1 door has car, 2 door have goat.

(a) $P\left(\text{your first choice is car}\right) = \frac{1}{3}$

if you stay of this,
$P(\text{win}) = \frac{1}{3}$

(b) your choice can be incorrect with $p = 2/3$.
$P(\text{win by switch}) = \frac{2}{3}$

(c) the switching helps you too double the probab. of winning.

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4. $M = \max(X, Y)$

$X \quad Y \quad M \in \{1, 2, 3, 4, 5, 6\}$

$P(M \le m) = P(X \le m, Y \le m)$
$= P(X \le m)\, P(Y \le m)$ by indep.

$\therefore P(X \le m) = \frac{m}{6}$

$\therefore P(M \le m) = \frac{m^2}{36}$

$P(M = m) = \frac{m^2}{36} - \frac{(m-1)^2}{36}$
$= \frac{\cancel{m^2} - \cancel{m^2} - 1 + 2m}{36} = \frac{2m-1}{36}$

m	1	2	3	4	5	6
$P(M=m)$	$1/36$	$3/36$	$5/36$	$7/36$	$9/36$	$11/36$

check ⇒ $\sum P(M = m) = 1$ $\left(\frac{1+3+5+7+9+11}{36}\right)$ 4 9 16 25 36

$= \frac{36}{36} = 1$

$E[M] = \sum_{i=1}^{6} m\, P(M = m)$

$= \frac{1}{36} + \frac{6}{36} + \frac{15}{36} + \frac{28}{36} + \frac{45}{36} + \frac{66}{36} = \frac{161}{36}$ 7 22 50 95

$E[M] = 4.4722$

5. $n = 120$ $X \sim$ (six appeared)

(a) $E[X] = np = 120 \times \frac{1}{6} = 20$

(b) $Var(X) = np(1-p) = \cancel{120}^{20} \times \frac{1}{\cancel{6}} \times \frac{5}{6} = \frac{100}{6} = 16.66$

(c) $SD(X) = \sqrt{Var(X)} = 4.08$

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6. $E[X] = 50$, $Var(X) = 64$

$SD(X) = 8$

chebyshev inequality $\boxed{P(|X-\mu| \geq k) \leq \frac{\sigma^2}{k^2}}$

(a) $P(|X-50| \geq 8) \Rightarrow \leq \frac{\sigma^2}{k^2} \leq \frac{\cancel{64}}{\cancel{64}} \leq 1$

(b) $P(|X-50| \geq 16) \Rightarrow \leq \frac{\cancel{64}^{4}}{\cancel{16}\times\cancel{16}\,4} \leq \frac{1}{4}$

(c) $P(34 \leq X \leq 66) \Rightarrow P(|X-50| \leq 16)$

$34 = 50 - 16$

$64 = 50 + 16$ $\qquad = 1 - \frac{1}{4} \geq \frac{3}{4}$

as $k\uparrow$, $k^2\uparrow$, so the upper bound increases. Therefore, guaranteed probab of being inside the interval increases.

7. $X_1 \dots X_n$ iid $\quad E[X_i] = \mu$, $Var(X_i) = \sigma^2$

$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$

(a) $E[\bar{X}] = E\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n} E(X_i)$

$= \frac{1}{\cancel{n}}\cancel{n}(\mu) = \mu$

(b) $Var(\bar{X}) = Var\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2} Var\left(\sum_{i=1}^{} X_i\right)$

$= \frac{1}{n^2}\, n\sigma^2 = \frac{\sigma^2}{n}$

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(c) $SD = \sqrt{\frac{\sigma^2}{n}} = \frac{\sigma}{\sqrt{n}}$

as $n \uparrow$, $\sqrt{n} \uparrow$, $\therefore$, $SD \downarrow$

(d) $\sigma = 12$,

$SD_{n=9} = \frac{12}{\sqrt{9}} = \frac{\cancel{12}}{\cancel{3}} = 4$

$SD_{n=36} = \frac{12}{\sqrt{36}} = \frac{\cancel{12}}{\cancel{6}} = 2$

$SD_{n=144} = \frac{12}{\sqrt{144}} = \frac{\cancel{12}}{\cancel{12}} = 1$

8. $X \sim N(162, 36)$ population $\mu = 162$, $\sigma^2 = 36$, $\sigma = 6$

$\therefore$

(a) $P(X > 170)$ $Z = \frac{X - \mu}{\sigma}$, $Z \sim N(0, 1)$

$P\left(Z > \frac{170-162}{6}\right) = P\left(Z > \frac{8}{6}\right) = 1 - \phi(1.33) =$
$= 0.0912$

(b) $P(156 < X < 168) = P\left(\frac{156-162}{6} < Z < \frac{168-162}{6}\right)$

$P(-1 < Z < 1) = \phi(1) - \phi(-1)$

$= 0.6826$

(c) $P(X \leq x) = 0.9$
$P\left(Z < \frac{x-162}{6}\right) = 0.9$ $\phi(z) = 0.9$
$= 1.2816$

~~P(~~ solve for x,
$\frac{x-162}{6} = 1.2816 \Rightarrow x = 169\ cm$

9. $X \sim N(\mu, \sigma^2)$

(a) $P(\mu - \sigma \leq X \leq \mu + \sigma) =$

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$$P\left(\frac{\cancel{\mu}-\sigma-\cancel{\mu}}{\sigma} < \frac{X-\mu}{\sigma} < \frac{\cancel{\mu}+\sigma-\cancel{\mu}}{\sigma}\right)$$

$$P(-1 < Z < 1) \qquad Z \sim N(0,1)$$

(b) $P(-1 < Z < 1) = \phi(1) - \phi(-1)$
$= 0.68268$

$P(-2 < Z < 2) = \phi(2) - \phi(-2)$
$= 0.95450$

$P(-3 < Z < 3) = \phi(3) - \phi(-3)$
$= 0.9973$

(c) $X \to Z\left(\frac{X-\mu}{\sigma}\right) \Rightarrow P(-k \leq Z < k)$
$\therefore$ these probabilities in b are universal for every normal distribu^n.

10. $X \sim N(\mu, \sigma^2)$ $\qquad \phi(z) = P(Z \leq x)$

(a) $P(X > \mu) = P(Z > 0) = \frac{1}{2}$

$P(X > \mu) = 1 - \phi(0) = \frac{1}{2}$

(b) $P(X < \mu + 2\sigma) = P\left(Z < \frac{\cancel{\mu}+2\sigma-\cancel{\mu}}{\sigma}\right)$
$= P(Z < 2) = \phi(2)$

(c) $P(\mu - \sigma < X < \mu + 2\sigma) = P(-1 < X < 2)$
$= \phi(2) - \phi(-1)$

Remember rule of symmetry,

$$\boxed{1 - \phi(t) = \phi(-t)}$$